

Bandit Algorithms

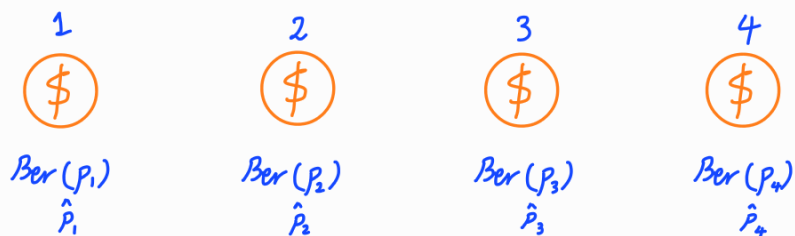
A major challenge in reinforcement learning is the exploration-exploitation trade-off. Bandit algorithms provide a rich framework to design and implement strategic exploration, especially in the online learning setting.

Regret: A measure of performance in online learning is regret, which is the difference between the learning agent's return and the return by the best policy.

instantaneous regret $\leftarrow$ regret = return (optimal policy) - return (learner's policy)
cumulative regret = $\sum_{t=1}^T$ return (optimal policy) - $\sum_{t=1}^T$ return (learner's policy at t)
 $\longrightarrow$ The goal is to minimize the agent's regret.

Multi-armed bandits $\longrightarrow$ Contextual bandits $\longrightarrow$ Markov decision process
ETC $\longrightarrow$ UCRL
UCB $\longrightarrow$ UCB-VI

Example:

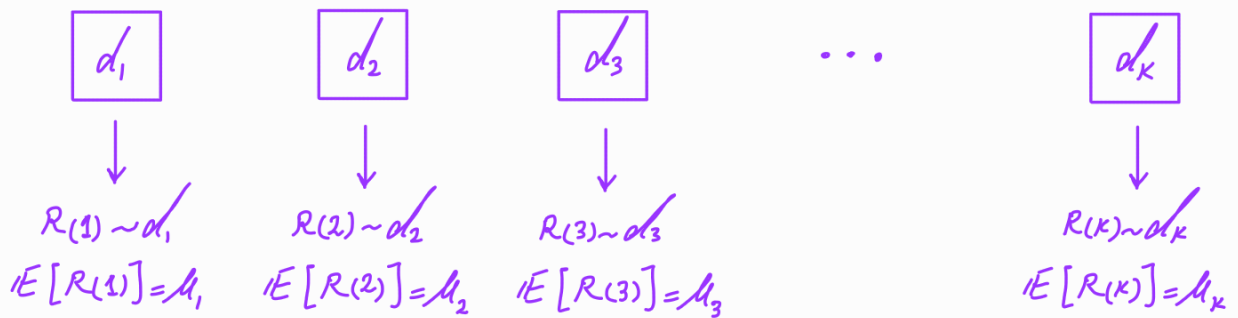


After T time steps: actual expected return = $\mathbb{E} [R_{t=1} + R_{t=2} + \dots + R_{t=T}] = \sum_{t=1}^T \mathcal{M}_{A_t}$
optimal expected return = $\mathbb{E} [R^* + R^* + \dots + R^*] = T \mathcal{M}^*$
cumulative regret = $T \mathcal{M}^* - \sum_{t=1}^T \mathcal{M}_{A_t}$

Multi-armed Bandit

Multi-armed bandit (MAB) is a special, simple class of RL problems with one state, i.e., $|S| = 1$. We will focus on the setting with finite actions (arms), i.e., $|A| = K$, and stochastic rewards, i.e.,

$R: A \rightarrow \Delta(\mathcal{R})$, that are unknown.



Agent's interaction with the environment:

- The interaction starts at time step $t=1$ (and state $s=1$).
- At time step $t \in \mathcal{N}$ (and state $s=1$), the agent takes action $A_t \in A$ (, observes the next state $s=1$,) and receives the immediate reward $R_t \sim R(A_t) \in \mathcal{R}$.
- The history of interaction at time t is

$$(A_1, R_1, A_2, R_2, \dots, A_t, R_t) \rightarrow \pi_{t+1}$$

Policy: A possibly randomized mapping from history to actions

$$\pi_t(\cdot \mid A_1, R_1, A_2, R_2, \dots, A_{t-1}, R_{t-1})$$

Goal: The learning agent aims to find a policy π that maximizes the expected cumulative reward

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\substack{A_t \sim \pi_t \\ R_t \sim R(A_t)}} \left[\sum_{t=1}^T R_t \right],$$

or equivalently, minimize the expected regret.

Note: If the actions are selected to be $A_1, A_2, A_3, \dots, A_T$, then we can rewrite the objective as

$$\text{Reg}(T) = \max_{A \in \mathcal{A}} \mathbb{E}_{R_t \sim R(A)} \left[\sum_{t=1}^T R_t \right] - \mathbb{E}_{\substack{A_t \sim \pi_t \\ R_t \sim R(A_t)}} \left[\sum_{t=1}^T R_t \right]$$

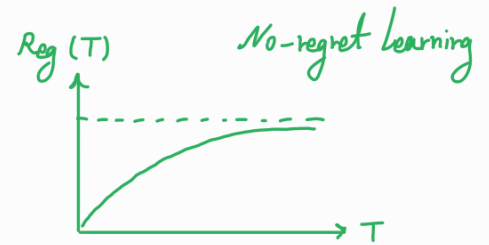
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The expected regret of the learning agent is

$$\text{Reg}(T) = T \mu^* - \sum_{t=1}^T \mu_{A_t}$$

* In online learning, a sublinear (in time horizon) regret is often desired:

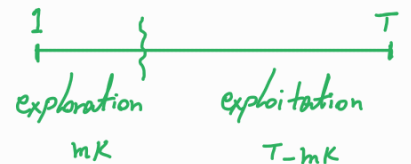
$$\lim_{T \rightarrow \infty} \frac{\text{Reg}(T)}{T} = 0$$



Explore-then-Commit algorithm: One can split the interaction horizon into

- exploration phase $\rightarrow$ exploring the arms to find the one with highest reward
- exploitation phase $\rightarrow$ committing to the arm thought to be the best

ETC Algorithm



- exploration** {
- For $t = 1, 2, \dots, mK$:
 - Pull arm (take action) $A_t = (t \bmod K)$
 - and observe reward $R_t \sim R(A_t)$
 - Compute the estimated mean $\hat{\mu}_a$ for all $a \in \mathcal{A}$

$$\hat{\mu}_a = \frac{1}{m} \sum_{t=1}^{mK} \mathbb{1}[A_t = a] R_t$$

- exploitation** {
- For $t = mK+1, mK+2, \dots, T$:
 - Pull the arm with the highest estimated mean
- $$A_t = \arg \max_{a \in \mathcal{A}} \hat{\mu}_a$$

Theorem: In the stochastic k -armed bandit with bounded rewards $R_t \in [0, 1]$ following arbitrary distributions, the following upper bound on the expected regret of the ETC algorithm holds:

$$\text{Reg}(T) \leq O\left(T^{\frac{2}{3}} k^{\frac{1}{3}} \left(\log \frac{k}{\delta}\right)^{\frac{1}{3}}\right)$$

with probability $1 - \delta$ for any $\delta \in (0, 1)$ if

$$m = \left(\frac{T \sqrt{\log(k/\delta)}}{2k} \right)^{\frac{2}{3}}.$$

Proof: We can decompose the regret into the regret during exploration and the regret during exploitation.

$$\begin{aligned} \text{Reg}(T) &= \mathbb{E} \left[\sum_{t=1}^T R(A^*) - R(A_t) \right] \\ &= \mathbb{E} \left[\sum_{t=1}^{mK} R(A^*) - R(A_{t \bmod k}) \right] && \text{exploration phase} \\ &\quad + \mathbb{E} \left[\sum_{t=mK+1}^T R(A^*) - R(\hat{A}^*) \right] && \text{exploitation phase} \end{aligned}$$

- Exploration phase: At most $K-1$ actions are suboptimal.

$$\mathbb{E} \left[\sum_{t=1}^{mK} R(A^*) - R(A_{t \bmod K}) \right] = \sum_{t=1}^{mK} \mu^* - \mu_{t \bmod K} \\ \leq m(K-1) \leq mK$$

- Exploitation phase: Let $\hat{A}^*$ be the action with the highest estimated mean

$$\hat{A}^* = \underset{a \in A}{\operatorname{argmax}} \hat{\mu}_a$$

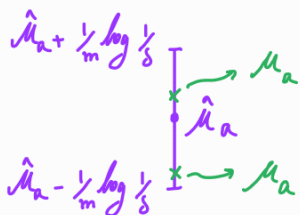
that will be taken in this phase.

$$\mathbb{E} \left[\sum_{t=mK+1}^T R(A^*) - R(\hat{A}^*) \right] = \sum_{t=mK+1}^T \mu^* - \mu_{\hat{A}^*} \\ = (T - mK) (\mu^* - \mu_{\hat{A}^*}) \leq T (\mu^* - \mu_{\hat{A}^*})$$

Using concentration inequalities, we can obtain probabilistic guarantees about how close the estimated means are to the true means.

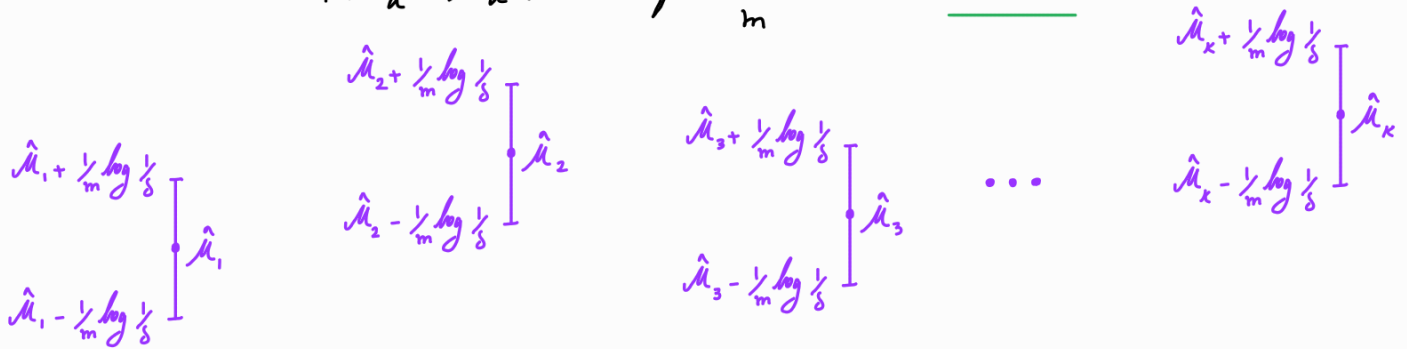
Based on Hoeffding's inequality, it holds that with probability $1 - \delta$,

$$|\hat{\mu}_a - \mu_a| \leq \sqrt{\frac{\log(\frac{2}{\delta})}{2m}} \stackrel{\delta \leq 0.5}{\leq} \sqrt{\frac{\log(\frac{1}{\delta})}{m}} \quad \text{for } a \in A$$



Next, based on union bound, it holds that with probability $1-\delta$,

$$|\hat{\mu}_a - \mu_a| \leq \sqrt{\frac{\log(\frac{K}{\delta})}{m}} \quad \forall a \in \mathcal{A}$$



Now, we can bound $\mu^* - \mu_{\hat{A}^*}$. We know that with probability $1-\delta$,

$$\begin{aligned} \mu_{A^*} - \mu_{\hat{A}^*} &\leq \left(\hat{\mu}_{A^*} + \sqrt{\frac{\log(\frac{K}{\delta})}{m}} \right) - \left(\hat{\mu}_{\hat{A}^*} - \sqrt{\frac{\log(\frac{K}{\delta})}{m}} \right) \\ &\leq \left(\hat{\mu}_{\hat{A}^*} + \sqrt{\frac{\log(\frac{K}{\delta})}{m}} \right) - \left(\hat{\mu}_{\hat{A}^*} - \sqrt{\frac{\log(\frac{K}{\delta})}{m}} \right) \\ &= 2\sqrt{\frac{\log(\frac{K}{\delta})}{m}} \end{aligned}$$

Therefore,

$$\mathbb{E} \left[\sum_{t=mK+1}^T R(A^*) - R(\hat{A}^*) \right] \leq 2T\sqrt{\frac{\log(\frac{K}{\delta})}{m}} \quad \text{w.p. } 1-\delta.$$

• Total regret:

$$\text{Reg}(T) \leq mK + 2T\sqrt{\frac{\log(\frac{K}{\delta})}{m}} \quad \text{w.p. } 1-\delta.$$

Minimizing the regret upper-bound, we obtain $m = \left(\frac{T\sqrt{\log(K/\delta)}}{2K} \right)^{2/3}$

and $\text{Reg}(T) \leq O\left(T^{2/3} K^{1/3} \left(\log \frac{K}{\delta}\right)^{1/3}\right) \quad \text{w.p. } 1-\delta.$

Upper Confidence Bound algorithm: Relying on the principle of optimism in the face of uncertainty, one can decide among arms based on an over-estimation of their mean reward. As a result, if a selected arm is

- near-optimal $\longrightarrow$ incur low regret
- far from optimal $\longrightarrow$ improve estimate

The UCB algorithm does not separate exploration and exploitation.

* At any time $t \in \mathbb{N}$, let $T_a(t)$ denote the number of times arm a has been pulled before, i.e.,

$$T_a(t) = \sum_{t'=1}^{t-1} \mathbb{1}[A_{t'} = a].$$

$$\sum_{a \in \mathcal{A}} T_a(t) = t-1$$

* At any time $t \in \mathbb{N}$, let $\hat{\mu}_a(t)$ denote the estimated mean reward of arm a based on the previous samples, i.e.,

$$\hat{\mu}_a(t) = \frac{1}{T_a(t)} \sum_{t'=1}^{t-1} \mathbb{1}[A_{t'} = a] R_{t'}.$$

$\longrightarrow$ with probability $1 - \delta$,

$$\hat{\mu}_a \pm 2 \sqrt{\frac{\log \frac{KT}{\delta}}{T_a(t)}} \quad \begin{matrix} \xrightarrow{\mu_a} \\ \xrightarrow{\mu_a} \end{matrix}$$

$$|\hat{\mu}_a(t) - \mu_a| \leq 2 \sqrt{\frac{\log \frac{KT}{\delta}}{T_a(t)}} \quad \forall a \in \mathcal{A}$$

UCB Algorithm

• Initialize $\hat{\mu}_a(1) = 0$ and $T_a(1) = 0$ for all $a \in \mathcal{A}$ → $t=1$

• For $t = 1, 2, \dots, T$:

• Pull the arm with the highest UCB index

$$A_t = \arg \max_{a \in \mathcal{A}} \hat{\mu}_a(t) + \sqrt{\frac{\log K T}{T_a(t)}}$$



• Update the number of samples for the selected arm

$$T_a(t+1) = \begin{cases} T_a(t) & \text{if } a \neq A_t \\ T_a(t) + 1 & \text{if } a = A_t \end{cases}$$

• Update the estimated mean of the selected arm

$$\hat{\mu}_a(t+1) = \begin{cases} \hat{\mu}_a(t) & \text{if } a \neq A_t \\ \hat{\mu}_a(t) + \frac{r_t - \hat{\mu}_a(t)}{T_a(t+1)} & \text{if } a = A_t \end{cases}$$

Theorem: In the stochastic k -armed bandit with bounded rewards
 sub-Gaussian $\leftarrow R_t \in [0, 1]$ following arbitrary distributions, the following
 upper bound on the expected regret of the UCB algorithm
 holds:

$$\text{Reg}(T) \leq O\left(\sqrt{KT \log \frac{KT}{\delta}}\right)$$

with probability $1 - \delta$ for any $\delta \in (0, 1)$.

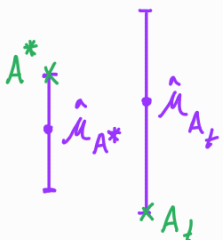
Proof: Recall that the regret is

$$\begin{aligned} \text{Reg}(T) &= \mathbb{E} \left[\sum_{t=1}^T R(A^*) - R(A_t) \right] \\ &= \sum_{t=1}^T \mu^* - \mu_{A_t} = T\mu^* - \sum_{t=1}^T \mu_{A_t} \end{aligned}$$

we have that at any time t with probability $1 - \delta$,

$$\mu^* - \mu_{A_t} \leq \left(\hat{\mu}_{A^*} + \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A^*}(t)}} \right) - \left(\hat{\mu}_{A_t} - \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A_t}(t)}} \right)$$

$$\begin{aligned} \text{UCB}_{A_t} \geq \text{UCB}_{A^*} &\leftarrow \leq \left(\hat{\mu}_{A_t} + \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A_t}(t)}} \right) - \left(\hat{\mu}_{A_t} - \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A_t}(t)}} \right) \\ &= 2 \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A_t}(t)}}. \end{aligned}$$



Therefore, with probability $1 - \delta$,

$$\begin{aligned} \text{Reg}(T) &\leq \sum_{t=1}^T 2 \sqrt{\frac{\log \frac{KT}{\delta}}{T_{A_t}(t)}} \\ &= 2 \sqrt{\log \frac{KT}{\delta}} \sum_{t=1}^T \sqrt{\frac{1}{T_{A_t}(t)}} \end{aligned}$$

We can show that

$$\begin{aligned} \sum_{t=1}^T \sqrt{\frac{1}{T_{A_t}(t)}} &= \sum_{t=1}^T \sum_{a \in A} \mathbb{1}[A_t = a] \sqrt{\frac{1}{T_a(t)}} \\ &= \sum_{a \in A} \sum_{t=1}^T \mathbb{1}[A_t = a] \sqrt{\frac{1}{T_a(t)}} \rightarrow \sqrt{\frac{1}{1}} + \sqrt{\frac{1}{2}} \\ &\quad + \dots + \sqrt{\frac{1}{T_a(T)}} \\ &= \sum_{a \in A} \sum_{m=1}^{T_a(T)} \sqrt{\frac{1}{m}} \end{aligned}$$

$$f(x) = \frac{1}{\sqrt{x}}$$

$$\sum_{m=1}^M \sqrt{\frac{1}{m}} \leq \int_0^M \sqrt{\frac{1}{m}} dm = 2\sqrt{m} \Big|_0^M = 2\sqrt{M} \leftarrow \leq \sum_{a \in A} 2\sqrt{T_a(T)}$$

$$\text{Jensen's inequality} \leftarrow \leq 2K \sqrt{\frac{1}{K} \sum_{a \in A} T_a(T)}$$

$$\sum_{a \in A} T_a(T) = T \leftarrow = 2K \sqrt{\frac{T}{K}} = 2\sqrt{KT}.$$

Finally, we have that with probability $1 - \delta$,

$$\text{Reg}(T) \leq 4 \sqrt{\log \frac{KT}{\delta}} \sqrt{KT}$$

and

$$\text{Reg}(T) \leq O\left(\sqrt{KT \log \frac{KT}{\delta}}\right).$$